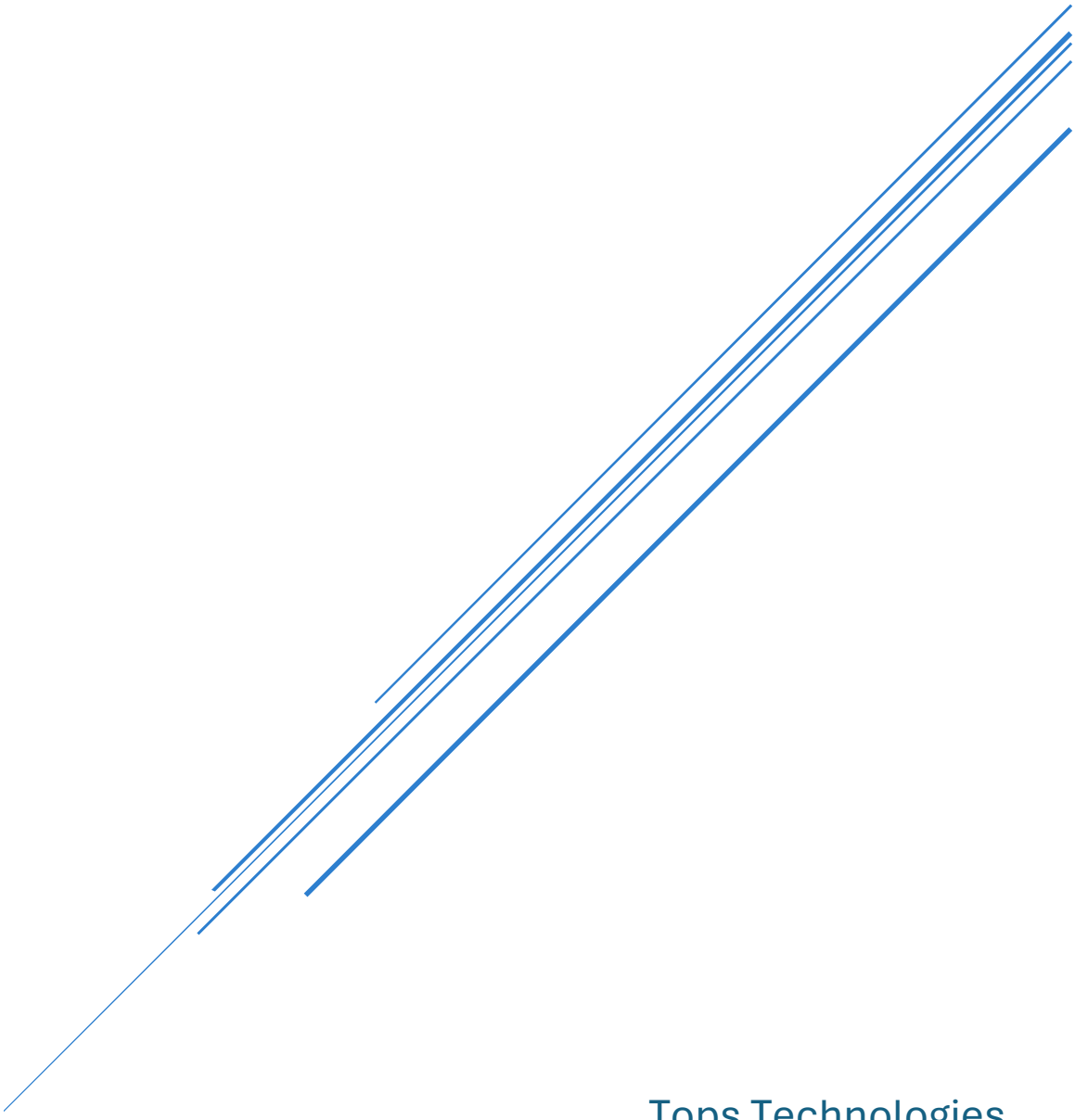
IP ROUTING AND ROUTING PROTOCOLS

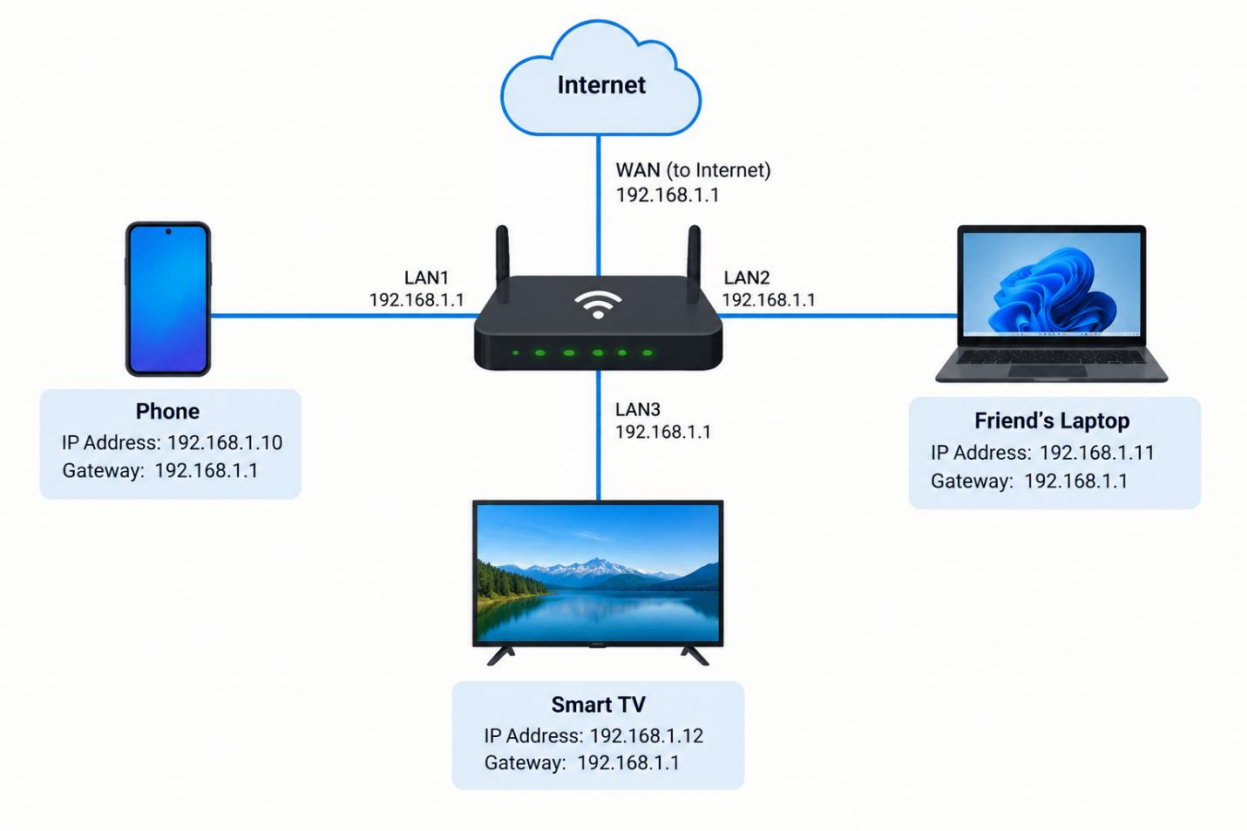
Assignment - 8



Tops Technologies
Cyber Security

1. **Draw a simple network diagram showing three devices (like your phone, a friend's laptop, and a smart TV) connected through a router, and label the IP addresses for each device and the router's interfaces.**

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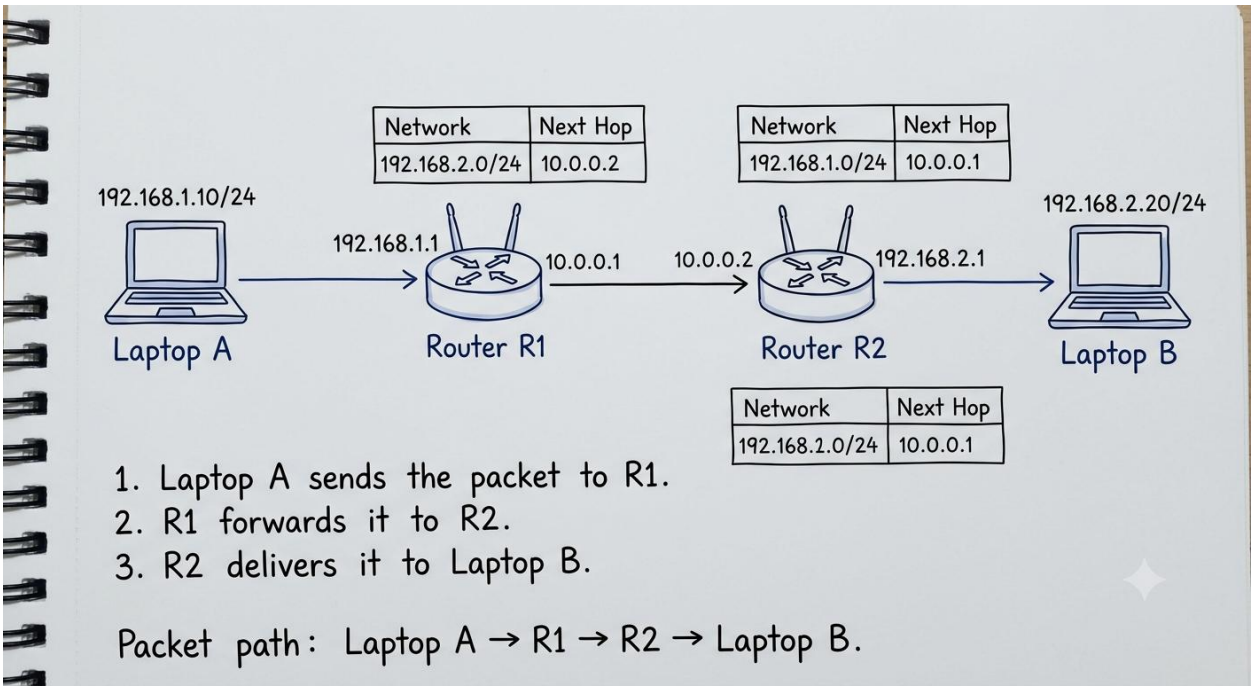


IP addressing used:

- **Phone: 192.168.1.10**
- **Friend's Laptop: 192.168.1.11**
- **Smart TV: 192.168.1.12**
- **Router LAN / Default Gateway: 192.168.1.1**

2. **Simulate manual IP routing by writing a step-by-step explanation of how a packet would travel from your laptop to a friend's computer in a different subnet, including which router(s) it passes through and how the routing decision is made.**
   Hint: Use made-up IP addresses and show the routing table entries that would be checked at each hop.

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Step 1: My laptop creates a packet.

My laptop has IP address 192.168.1.10, and my friend's laptop has IP address 192.168.2.20. Since they are in different subnets, my laptop sends the packet to its default gateway, 192.168.1.1.

Step 2: Router R1 checks its routing table.

Router R1 checks the destination IP address 192.168.2.20 and selects the route toward Router R2.

Destination network	Next hop	Interface
192.168.1.0/24	Directly connected	LAN
10.0.0.0/30	Directly connected	Link to R2
192.168.2.0/24	10.0.0.2	Link to R2

Step 3: Router R1 forwards the packet.

R1 forwards the packet to R2 at 10.0.0.2. The packet's destination IP remains 192.168.2.20.

Step 4: Router R2 checks its routing table.

R2 finds that the destination network 192.168.2.0/24 is directly connected to its LAN interface.

Destination network	Next hop	Interface
10.0.0.0/30	Directly connected	Link to R1
192.168.2.0/24	Directly connected	LAN

Step 5: Router R2 delivers the packet.

R2 forwards the packet to my friend's laptop at 192.168.2.20 on the local network.

Step 6: My friend's laptop receives the packet.

The destination laptop receives the packet. If a reply is sent, it uses its own routing table and default gateway to return the traffic.

3. **Create a routing table for a fictional Zomato-style delivery network with three branches in different cities, each on a different subnet. Specify the network address, subnet mask, next hop, and interface for each route.**

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Routing table for Router R1 (Ahmedabad)

Network address	Subnet mask	Next hop	Interface
192.168.1.0	255.255.255.0	Directly connected	G0/0
192.168.2.0	255.255.255.0	10.0.12.2	G0/1
192.168.3.0	255.255.255.0	10.0.12.2	G0/1

Routing table for Router R2 (Mumbai)

Network address	Subnet mask	Next hop	Interface
192.168.1.0	255.255.255.0	10.0.12.1	G0/1
192.168.2.0	255.255.255.0	Directly connected	G0/0
192.168.3.0	255.255.255.0	10.0.23.2	G0/2

Routing table for Router R3 (Pune)

Network address	Subnet mask	Next hop	Interface
192.168.1.0	255.255.255.0	10.0.23.1	G0/1
192.168.2.0	255.255.255.0	10.0.23.1	G0/1
192.168.3.0	255.255.255.0	Directly connected	G0/0

How routing works

1. **A packet from Ahmedabad to Mumbai reaches Router R1.**
2. **R1 checks its routing table and forwards the packet to the next hop, 10.0.12.2.**
3. **Router R2 recognizes Mumbai's network as directly connected and delivers the packet.**
4. **For Pune, R2 forwards packets to 10.0.23.2, which represents Router R3.**

4. **Compare static routing and dynamic routing protocols (like RIP or OSPF) by listing 3 pros and 3 cons of each, using examples from real-world apps or services you use (e.g., how a food delivery app might benefit from dynamic routing).**

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Type	Pros (Advantages)	Cons (Disadvantages)	Real-world example
Static Routing	1. Easy to configure in small networks. 2. Uses fewer router resources. 3. More predictable and controlled.	1. Routes must be updated manually. 2. Does not automatically adapt to link failures. 3. Difficult to manage in large networks.	A small restaurant uses a fixed network to connect its billing system and printer.
Dynamic Routing (RIP)	1. Automatically learns routes. 2. Updates routes when network changes are detected. 3. Easy to configure for small networks.	1. RIP has a maximum of 15 hops. 2. Convergence can be slow. 3. Chooses routes mainly by hop count, not bandwidth.	A small food delivery company's branch network uses RIP to exchange routes between offices.
Dynamic Routing (OSPF)	1. Quickly adapts to many network failures. 2. Supports large and complex networks. 3. Selects routes using link cost, often based on bandwidth.	1. More complex to configure. 2. Uses more CPU and memory than simple static routing. 3. Requires proper planning and troubleshooting skills.	A large delivery service connects its servers and city offices using OSPF to find alternative paths if a link fails.

5. **Use ChatGPT or another AI tool to generate a sample configuration for setting up OSPF routing between two routers, then explain in your own words what each configuration line does.**
   Hint: Paste the AI-generated config and annotate each part with a short description.

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OSPF Configuration Between Two Routers

We will configure OSPF (Open Shortest Path First) between two Cisco routers, R1 and R2, so they can exchange routing information automatically.

Network diagram

PC1 — Network 1

192.168.1.10/24

Router R1

192.168.1.1

Connection: 10.0.0.1 ↔ 10.0.0.2

Router R2

192.168.2.1

PC2 — Network 2

192.168.2.10/24

1. Configuration for Router R1

Cisco IOS commands

```
enable
configure terminal
interface gigabitEthernet 0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```

exit

interface gigabitEthernet 0/1

ip address 10.0.0.1 255.255.255.252

no shutdown

exit

router ospf 1

network 192.168.1.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

end

2. Configuration for Router R2

enable

configure terminal

interface gigabitEthernet 0/0

ip address 192.168.2.1 255.255.255.0

no shutdown

exit

interface gigabitEthernet 0/1

ip address 10.0.0.2 255.255.255.252

no shutdown

exit

router ospf 1

network 192.168.2.0 0.0.0.255 area 0

network 10.0.0.0 0.0.0.3 area 0

end

Note: The interface names assume each router has GigabitEthernet 0/0 and 0/1. Adjust them if your Packet Tracer router uses different names.

3. Explanation of configuration commands

Command	Explanation
enable	Enters privileged EXEC mode.
configure terminal	Enters global configuration mode.
interface gigabitEthernet 0/0	Selects the specified interface.
ip address ...	Assigns an IP address and subnet mask.
no shutdown	Enables the interface.
exit	Returns to the previous configuration mode.
router ospf 1	Enables OSPF process 1.
network ... area 0	Enables OSPF on matching interfaces and advertises their networks in Area 0.
end	Returns to privileged EXEC mode.

What is a wildcard mask?

- **0.0.0.255 matches the /24 network.**
- **0.0.0.3 matches the /30 router-to-router network.**

4. Verify the configuration

Run these commands on both routers:

show ip ospf neighbor

show ip route ospf

show ip protocols

- **show ip ospf neighbor checks whether the routers have formed an OSPF neighbor relationship.**

- **show ip route ospf displays routes learned through OSPF.**
- **show ip protocols displays routing protocol information.**